Exercise 2: Continuum Mechanics

Task 2.1: Mass Balance

Due to the incompressibility of the fluid, the density ρ is constant over time. In addition, the geometry of the pipe does not change, so that the surface element $\mathrm{d}a$ of the current configuration matches that of the reference configuration ($\mathrm{d}A$). The global form of the mass balance is therefore simplified to

$$\int_{\partial\mathcal{B}} \rho\, \dot{\boldsymbol{x}} \cdot \boldsymbol{n}\, \mathrm{d}A = 0.$$

The integration over the boundary of the area $\mathcal{B}$ can be carried out separately for the three boundary surfaces A^{I} to A^{III}, which are characterized by the outwardly directed normal vectors $\boldsymbol{n}^{\mathrm{I}}$ to $\boldsymbol{n}^{\mathrm{III}}$, see Fig. 2.1.

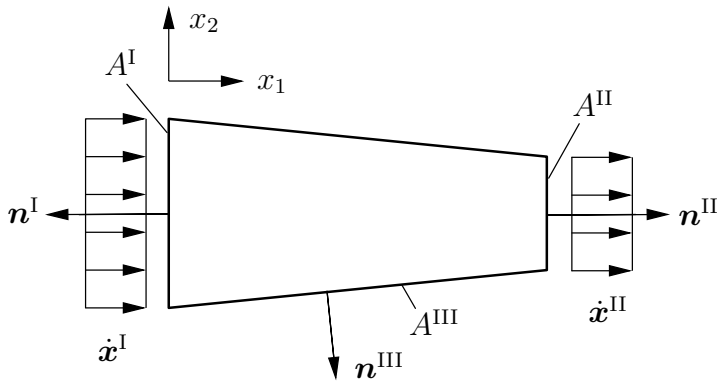


Figure 2.1: Labeling of the surfaces and their normal vectors.

At the lateral surface of the pipe, the fluid is forced to flow parallel to the pipe wall. The velocity vectors are therefore perpendicular to the normal vectors and the scalar product disappears,

$$\dot{\boldsymbol{x}}^{\mathrm{III}} \cdot \boldsymbol{n}^{\mathrm{III}} = 0.$$

The mass balance thus leads to the condition

$$0 = \int_{A^{\mathrm{I}}} \rho\, \dot{\boldsymbol{x}}^{\mathrm{I}} \cdot \boldsymbol{n}^{\mathrm{I}}\, \mathrm{d}A + \int_{A^{\mathrm{II}}} \rho\, \dot{\boldsymbol{x}}^{\mathrm{II}} \cdot \boldsymbol{n}^{\mathrm{II}}\, \mathrm{d}A.$$

With the normal vectors $\boldsymbol{n}^{\mathrm{I}} = \begin{bmatrix} -1 & 0 \end{bmatrix}^T$ and $\boldsymbol{n}^{\mathrm{II}} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T$ follows

$$0 = -\int_{A^{\mathrm{I}}} \rho\, \dot{x}_1^{\mathrm{I}}\, \mathrm{d}A + \int_{A^{\mathrm{II}}} \rho\, \dot{x}_1^{\mathrm{II}}\, \mathrm{d}A$$
$$0 = -\rho\, \dot{x}_1^{\mathrm{I}} A^{\mathrm{I}} + \rho\, \dot{x}_1^{\mathrm{II}} A^{\mathrm{II}}.$$

The continuity equation is obtained for a constant flow rate

$$Q := \dot{x}_1^{\mathrm{I}} A^{\mathrm{I}} = \dot{x}_1^{\mathrm{II}} A^{\mathrm{II}} = \text{const.}$$

Task 2.2: Deformation Gradient and Strain Measure

a) The deformation gradient is defined as $\boldsymbol{F} = \dfrac{\partial \boldsymbol{x}}{\partial \boldsymbol{X}}$ and thus has the components

$$F_{ij} = \begin{bmatrix} 1 & \alpha & 0 \\ 0 & 1 & \beta \\ \gamma & 0 & 1 \end{bmatrix}.$$

b) The coefficients of the deformed vectors $\boldsymbol{a}$, $\boldsymbol{b}$ and $\boldsymbol{c}$ can be calculated by multiplying the coefficient matrix of $\boldsymbol{F}$ with the column matrices $\begin{bmatrix} 1 & 0 & 0 \end{bmatrix}^T$, $\begin{bmatrix} 0 & 1 & 0 \end{bmatrix}^T$ and $\begin{bmatrix} 0 & 0 & 1 \end{bmatrix}^T$. Alternatively, the index notation can be used, in which the components of $\boldsymbol{a} = \boldsymbol{F}\boldsymbol{A}$ are given by $a_i = F_{ij}A_j$. From the full form $a_i = F_{i1}A_1 + F_{i2}A_2 + F_{i3}A_3$ we obtain

$$\begin{aligned} a_1 &= 1 \cdot 1 + \alpha \cdot 0 + 0 \cdot 0 = 1, \\ a_2 &= 0 \cdot 1 + 1 \cdot 0 + \beta \cdot 0 = 0, \\ a_3 &= \gamma \cdot 1 + 0 \cdot 0 + 1 \cdot 0 = \gamma. \end{aligned}$$

And in the component notation, the vector is represented as $\boldsymbol{a} = \boldsymbol{e}_1 + \gamma\, \boldsymbol{e}_3$. When calculating $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{B}$ directly rather than as $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{e}_2$, the resulting components are $b_i = F_{ij}\delta_{j2} = F_{i2}$ and we get

$$b_1 = F_{12} = \alpha, \qquad b_2 = F_{22} = 1, \qquad b_3 = F_{32} = 0$$

or $\boldsymbol{b} = \alpha\, \boldsymbol{e}_1 + \boldsymbol{e}_2$. Similarly, $\boldsymbol{c} = \beta\, \boldsymbol{e}_2 + \boldsymbol{e}_3$ is obtained from $c_i = F_{ij}\delta_{j3} = F_{i3}$. The deformation is shown for $\alpha = \gamma = 0$ in Fig. 2.2.

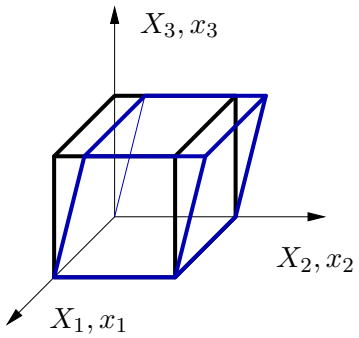


Figure 2.2: Sketch of the deformed unit cube for $\alpha = \gamma = 0$.

c) The volume v of the deformed body is calculated from the initial volume V using the relationship $v = \det[\boldsymbol{F}]\, V = J\, V$. For the general case, the following applies

$$J = 1 + \alpha\beta\gamma, \tag{2.1}$$

In the special case $\alpha = \gamma = 0$ there is no volume change due to $J = 1$.

d) In index notation, the components of the right Cauchy-Green strain tensor are $C_{ik} = F_{ij}^T F_{jk} = F_{ji} F_{jk}$. Thus follows

$$\begin{aligned}
C_{11} &= F_{j1}F_{j1} = F_{11}^2 + F_{21}^2 + F_{31}^2 = 1^2 + 0^2 + \gamma^2 = 1 + \gamma^2 \\
C_{21} &= F_{j2}F_{j1} = F_{12}F_{11} + F_{22}F_{21} + F_{32}F_{31} = \alpha \cdot 1 + 1 \cdot 0 + 0 \cdot \gamma = \alpha \\
&\vdots \\
C_{33} &= F_{j3}F_{j3} = F_{13}^2 + F_{23}^2 + F_{33}^2 = 0^2 + \beta^2 + 1^2 = \beta^2 + 1,
\end{aligned}$$

in matrix representation this corresponds to

$$C_{ij} = \begin{bmatrix} 1+\gamma^2 & \alpha & \gamma \\ \alpha & 1+\alpha^2 & \beta \\ \gamma & \beta & 1+\beta^2 \end{bmatrix}.$$

The same result is obtained when the coefficient matrices of $\boldsymbol{F}^T$ and $\boldsymbol{F}$ are multiplied:

$$C_{ij} = \begin{bmatrix} 1 & 0 & \gamma \\ \alpha & 1 & 0 \\ 0 & \beta & 1 \end{bmatrix} \begin{bmatrix} 1 & \alpha & 0 \\ 0 & 1 & \beta \\ \gamma & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1+\gamma^2 & \alpha & \gamma \\ \alpha & 1+\alpha^2 & \beta \\ \gamma & \beta & 1+\beta^2 \end{bmatrix}.$$

e) The components of the Green strain tensor $\boldsymbol{E} = \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I})$ are

$$E_{ij} = \tfrac{1}{2}\left(C_{ij} - \delta_{ij}\right) = \tfrac{1}{2} \begin{bmatrix} \gamma^2 & \alpha & \gamma \\ \alpha & \alpha^2 & \beta \\ \gamma & \beta & \beta^2 \end{bmatrix}. \tag{2.2}$$

If $\boldsymbol{E}$ is geometrically linearized, the infinitesimal strain tensor $\boldsymbol{\varepsilon}$ is obtained as

$$\boldsymbol{\varepsilon} = \tfrac{1}{2}\left(\mathrm{Grad}[\boldsymbol{u}] + \mathrm{Grad}^T[\boldsymbol{u}]\right).$$

To obtain a representation dependent on $\boldsymbol{F}$, $\boldsymbol{u}$ is replaced by $\boldsymbol{x} - \boldsymbol{X}$ and the result is as follows

$$\begin{aligned}
\boldsymbol{\varepsilon} &= \tfrac{1}{2}\left(\mathrm{Grad}[\boldsymbol{x}] - \boldsymbol{I} + \mathrm{Grad}^T[\boldsymbol{x}] - \boldsymbol{I}\right) \\
&= \tfrac{1}{2}\left(\mathrm{Grad}[\boldsymbol{x}] + \mathrm{Grad}^T[\boldsymbol{x}]\right) - \boldsymbol{I} \\
&= \tfrac{1}{2}\left(\boldsymbol{F} + \boldsymbol{F}^T\right) - \boldsymbol{I} \\
&= \mathrm{sym}[\boldsymbol{F}] - \boldsymbol{I}.
\end{aligned}$$

The coefficient matrix of the linearized strain tensor is calculated as follows

$$\varepsilon_{ij} = \tfrac{1}{2}\left(F_{ij} + F_{ji}\right) - \delta_{ij} = \tfrac{1}{2} \begin{bmatrix} 0 & \alpha & \gamma \\ \alpha & 0 & \beta \\ \gamma & \beta & 0 \end{bmatrix}.$$

A comparison with Eq. (2.2) shows that the main diagonal elements have disappeared as a result of the linearization. The volume expansion $\varepsilon_{\mathrm{V}} = \mathrm{tr}[\boldsymbol{\varepsilon}]$ now also results in $\alpha \neq 0, \gamma \neq 0$ in the case of $\varepsilon_{\mathrm{V}} = 0$. The prerequisite for the use of $\boldsymbol{\varepsilon}$ are small displacements $\boldsymbol{u}$, i. h.

$$x_1 \approx X_1, \quad x_2 \approx X_2, \quad x_3 \approx X_3,$$

or

$$u_1 = \alpha X_2 \approx 0, \quad u_2 = \beta X_3 \approx 0, \quad u_3 = \gamma X_1 \approx 0.$$

This is given for $\alpha \ll 1$, $\beta \ll 1$, $\gamma \ll 1$. Then it also follows from Eq. (2.1) with

$$J = 1 + \alpha\beta\gamma \approx 1,$$

that the volume does not change during deformation.

Task 2.3: Parameter determination

The Lamé constants λ and μ are identified from the given test data by inserting the known stress boundary conditions $\sigma_{33} = \sigma_{22} = 0$ and the determined relationships between σ_{11} and ε_{11} or ε_{22} and ε_{11} into the constitutive equations

$$\sigma_{11} = (\lambda + 2\mu)\,\varepsilon_{11} + \lambda\varepsilon_{22} + \lambda\varepsilon_{33} \tag{2.3}$$

$$\sigma_{22} = \lambda\varepsilon_{11} + (\lambda + 2\mu)\,\varepsilon_{22} + \lambda\varepsilon_{33} \tag{2.4}$$

$$\sigma_{33} = \lambda\varepsilon_{11} + \lambda\varepsilon_{22} + (\lambda + 2\mu)\,\varepsilon_{33}. \tag{2.5}$$

If we first set $\sigma_{33} = 0$, we obtain from Eq,(2.5)

$$\varepsilon_{33} = -\frac{\lambda}{\lambda + 2\mu}\,(\varepsilon_{11} + \varepsilon_{22}). \tag{2.6}$$

This and $\sigma_{22} = 0$ inserted in Eq. (2.4) leads to

$$\varepsilon_{22} = -\frac{\lambda}{2\,(\lambda + \mu)}\,\varepsilon_{11}, \tag{2.7}$$

which makes it possible to identify $2\,(\lambda + \mu)/\lambda$ from the right-hand diagram in Fig. **??**. Using Eq. (2.3) and substituting Eq. (2.7) and (2.6), the equation

$$\sigma_{11} = \frac{3\mu\lambda^2 + 8\mu^2\lambda + 4\mu^3}{(\lambda + \mu)(\lambda + 2\mu)}\,\varepsilon_{11} \tag{2.8}$$

can be obtained. Using Fig. **??**, the relationships

$$\frac{\varepsilon_{11}}{\varepsilon_{22}} \approx -5 \qquad \text{and} \qquad \frac{\sigma_{11}}{\varepsilon_{11}} \approx 150\,\text{GPa} \tag{2.9}$$

can be inferred (see Fig. 2.3). Inserting this into Eq. (2.7), we obtain $\mu \approx \frac{3}{2}\,\lambda$ and we can determine the value of λ by inserting the above results into Eq. (2.8) as follows

$$150\,\text{GPa} \approx \frac{\frac{9}{2} + 18 + \frac{27}{2}}{\frac{5}{2} \cdot 4}\,\lambda = \frac{18}{5}\,\lambda \qquad \Rightarrow \qquad \lambda \approx 42\,\text{GPa}, \quad \mu \approx 62.5\,\text{GPa}. \tag{2.10}$$

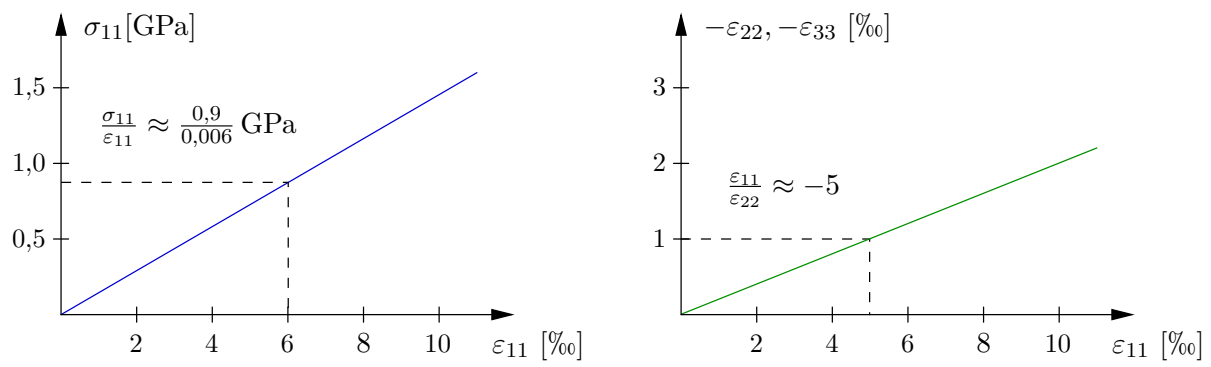


Figure 2.3: Reading the ratios $\sigma_{11}/\varepsilon_{11}$ and $\varepsilon_{11}/\varepsilon_{22}$.